

PACKIARAJAN SIVASUBRAMANIAN

GIS Intern

Chennai, India • p15rajan@gmail.com • +91 9886027591

PROFESSIONAL SUMMARY

Detail-oriented GIS professional currently pursuing an M. Tech in Geoinformatics (University of Madras) with hands-on expertise in ArcGIS Pro, QGIS, FME Workbench, and PostGIS. Combines a strong foundation in spatial analysis, remote sensing, and UAV data processing with 14+ years of enterprise IT experience in cloud infrastructure and data management. Adept at LULC classification, ETL workflows, and geospatial data visualization. Eager to apply geospatial intelligence and data-driven problem-solving skills in a challenging GIS internship environment.

EDUCATION

M.Tech in Geoinformatics <i>University of Madras, Chennai, India</i>	2025 – 2027 (Pursuing)
M.Sc. Data Science — CGPA: 7.1 Chandigarh University, India	2023 – 2025
B.Sc. Mathematics & Chemistry Madurai Kamaraj University, India	1981-1984

CORE COMPETENCIES

GIS Platforms: ArcGIS Pro, ArcMap, QGIS, SAGA	Remote Sensing: ENVI 5.6, Pix4D, UAV/Drone Data Processing
Spatial Analysis: Buffer, Overlay, Network, Neighbourhood, Spatial Join, Interpolation	Data Processing: Georeferencing, Digitization, Geoprocessing, Ortho Mosaic Creation
ETL & Workflows: FME Workbench Pipelines, Data Integration, Workflow Automation	Classification: LULC Classification, Biomass Estimation, Crop Yield Calculation
Databases: PostgreSQL, PostGIS, Neo4j (graph)	Programming: Python, SQL, R — data manipulation, spatial analysis, scripting
Data Formats: Shapefile, GeoJSON, KML, TIFF, Point Cloud, Raster/Vector	Cloud: AWS, Azure
Image Analysis: Multi-spectral & Hyperspectral, Photogrammetry	Standards: ISO 19115, Spatial Data Metadata Management

TECHNICAL CERTIFICATIONS

- Coursera** — Mastering Geospatial Analysis with QGIS
- Google / Coursera** — Data Analysis with R Programming
- Coursera** — Database Design and Basic SQL in PostgreSQL
- Udemy** — FME Essentials

KEY GIS PROJECTS & ACHIEVEMENTS

Geospatial Analysis: Global Warming Impact Assessment

2023 Tools: ArcGIS Pro, QGIS

- Performed comprehensive spatial analysis — Buffer, Overlay, and LULC classification — to quantify land-use change over time.
- Generated thematic heat maps and geospatial visualizations to illustrate and communicate climate change impacts to stakeholders.
- Produced actionable insights by correlating land surface temperature anomalies with land-cover change patterns.

Flood Risk Zonation — Cuddalore District, Tamil Nadu

2025 Tools: ArcGIS Pro, Python, arcpy

- Delineated flood-risk zones across Cuddalore district by integrating multi-criteria spatial datasets — DEM (SRTM/ASTER), LULC classification, soil permeability, slope, and historical rainfall records.
- Applied Weighted Overlay Analysis in ArcGIS Pro to generate a composite flood susceptibility index, classifying the district into Low, Moderate, High, and Very High-risk zones.
- Produced thematic risk maps and a stakeholder report recommendations.

Customer Churn Prediction — Spatial Business Analytics

2024 Tools: Python, Scikit-learn, Machine Learning

- Built binary classification models (Logistic Regression) achieving 81% accuracy and 0.85 F1-score on large customer datasets.
- Conducted exploratory data analysis (EDA) and feature engineering, demonstrating strong quantitative data handling applicable to geospatial datasets.

Traffic Signal Detection using Computer Vision

2023 Tools: TensorFlow, Keras, Python

- Designed a CNN-based model for real-world traffic signal detection — relevant to smart-city GIS and transportation analysis applications.
- Evaluated model performance using precision, recall, and cross-validation metrics for robust detection accuracy.

PROFESSIONAL EXPERIENCE

Senior Infrastructure Architect

Jan 2008 – May 2022 Cognizant Technology Solutions, India

- Led pre-sales activities including RFP/RFI responses, crafting tailored cloud infrastructure solutions for enterprise clients.
- Planned and designed Infrastructure-as-a-Service (IaaS) solutions across AWS and Azure, managing large-scale data environments — skills directly transferable to cloud-hosted GIS platforms.
- Managed end-to-end workload migration from on-premises systems to public cloud, ensuring data integrity and zero downtime.

Member Technical Staff / Storage Consultant

Jul 2003 – Jun 2006 Sunrise Systems Inc., USA & UBS Singapore

- Planned, designed, and implemented SAN, NAS, and enterprise backup systems, specializing in performance optimization and data protection. Ensured high availability, security, and efficiency of large-scale data environments supporting multi-terabyte workloads.

ADDITIONAL SKILLS

- Strong analytical and spatial problem-solving abilities with a geoscience mindset
- Effective data visualization and technical communication for non-technical audiences
- Experience with remote sensing data processing, interpretation, and multi-spectral image analysis
- Collaborative team player with cross-functional experience across IT and academia